Week 3 | **CLO** 1-A · 2-A · 3-B · 3-C | **Textbook** p. 178 {2, 3, 10, 17} | **File** 2 / 8

3-1 Measures of Central Tendency

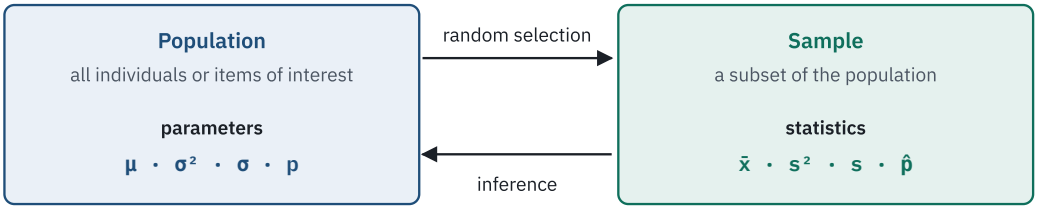
مقاييس النزعة المركزية — المتوسط والوسيط والمنوال ومنتصف المدى والمتوسط المرجّح

Chapter 2 organised data into frequency distributions and graphs. Chapter 3 **summarises data numerically** with three families of measures: **central tendency** (mean, median, mode, midrange, weighted mean), **variation** (range, variance, standard deviation) and **position** (z-scores, percentiles, quartiles, deciles, interquartile range). Together these make up *traditional statistics*.

HADHRAMI

0 · Parameter vs statistic, and the rounding rule

BOTH · CLO1-A



Parameters describe the whole population and use Greek letters; statistics describe a sample and use Latin letters. We select a sample at random, compute a statistic, and use it to infer the parameter.

المعلمة (parameter) تصف المجتمع كله وتُكتب بحروف يونانية، والإحصاءة (statistic) تصف العينة. نسحب عينة عشوائية، نحسب الإحصاءة، ثم نستدل بها على المعلمة.

A **parameter** is a numerical value that describes an **entire population**, computed from all its data values (e.g. μ, σ^2). A **statistic** is a descriptive measure computed from a **sample** (e.g. $\bar{x}$, s^2). Statisticians usually study populations through samples. When a population is **small**, no sample is needed: the whole population can be studied.

MEASURE	STATISTIC	PARAMETER
Mean	$\bar{x}$ (x bar)	μ (mu)
Variance	s^2	σ^2 (sigma squared)
Standard deviation	s	σ (sigma)
Proportion	$\hat{p}$ (p hat)	p

GENERAL ROUNDING RULE

Do **not** round during the calculation, only at the final answer. Carry three or four decimal places through the working. Example 3-1 gives $578/9 \approx 64.2$, but the slide marks 64.222 as the better working value.

الشرح بالعربي — المعلمة والإحصاءة

المعلمة (parameter) رقم يصف **المجتمع كله** ويُحسب من كل بياناته، ويُرمز له بحروف يونانية: μ للمتوسط و σ^2 للتباين. الإحصاءة (statistic) رقم يصف **العينة**، ويُرمز له بحروف لاتينية: $\bar{x}$ و s^2. سؤال شائع: «متوسط أعمار كل طلاب جامعة السلطان قابوس» ← معلمة. «متوسط أعمار 50 طالبًا اخترناهم» ← إحصاءة. وقاعدة التقريب: **لا تقرّب في منتصف الحل**، بل في الإجابة النهائية فقط.

1 · The mean

BOTH

A **measure of central tendency** summarises a whole data set with one value that represents its middle or centre. The **mean** (arithmetic average) is the sum of the values divided by the number of values. It applies to discrete and continuous data.

SAMPLE MEAN (A STATISTIC) · POPULATION MEAN (A PARAMETER)

$$\bar{x} = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{\Sigma X}{n} \qquad \mu = \frac{X_1 + X_2 + \cdots + X_N}{N} = \frac{\Sigma X}{N}$$

$\bar{x}$ is pronounced "X bar", μ "mew". n = number of values in the sample, N = number of values in the population.

EX 3-1 · AVIAN FLU CASES BOTH

Flu cases over 9 years: 4, 46, 98, 115, 88, 44, 73, 48, 62. Find the mean.

$\bar{x} = 578/9 = 64.222 \approx$ **64.2**

The mean number of flu cases over the 9 years is 64.2.

EX 3-2 · STORE SALES HADHRAMI

Sales (in $ millions) over 5 years: 221, 239, 262, 281, 318. Find the mean.

$\bar{x} = 1321/5 =$ **264.2**

The mean is $264.2 million, which is *not* one of the data values.

PROPERTIES OF THE MEAN · MEMORISE ALL SIX BOTH

(1) It uses **all** the data values. (2) It varies **less** than the median or mode across samples from the same population. (3) It is used to compute other statistics, such as the variance. (4) It is **unique** and usually **not** one of the data values. (5) It **cannot be computed** for a frequency distribution with an **open-ended class**. (6) It is strongly affected by **outliers** (extremely high or low values), so it may be the wrong average to report. Also, knowing the mean alone is not enough to describe a data set; see 3-2.

2 · The mean for grouped data BOTH

PROCEDURE: TABLE A CLASS · B FREQUENCY F · C MIDPOINT X_m · D $F \cdot X_m$

$$X_m = \frac{\text{lower} + \text{upper}}{2} \quad \text{(limits or boundaries give the same midpoint)} \qquad \bar{x} = \frac{\Sigma f \cdot X_m}{n}$$

Steps: find each midpoint, multiply by the frequency (column D), sum column D, then divide by n = Σf.

EX 3-3 · SALARIES OF CEOS ($ MILLIONS) HADHRAMI

CLASS	F	X_m	$F \cdot X_m$
15.5–20.5	13	18	234
20.5–25.5	6	23	138
25.5–30.5	4	28	112
30.5–35.5	1	33	33
35.5–40.5	1	38	38
Total	**25**		**555**

$\bar{x} = 555/25 =$ **$22.2 million**

AGES OF DRIVERS AMANI

LIMITS	BOUNDARIES	F	X_m	$F \cdot X_m$	CF
21–28	20.5–28.5	19	24.5	465.5	19
29–36	28.5–36.5	20	32.5	650	39
37–44	36.5–44.5	16	40.5	648	55
45–52	44.5–52.5	14	48.5	679	69
53–60	52.5–60.5	0	56.5	0	69
61–68	60.5–68.5	1	64.5	64.5	70
69–76	68.5–76.5	1	72.5	72.5	71
Total		**71**		**2579.5**	

$\bar{x} = 2579.5/71 = 36.331 \approx$ **36.3 years**

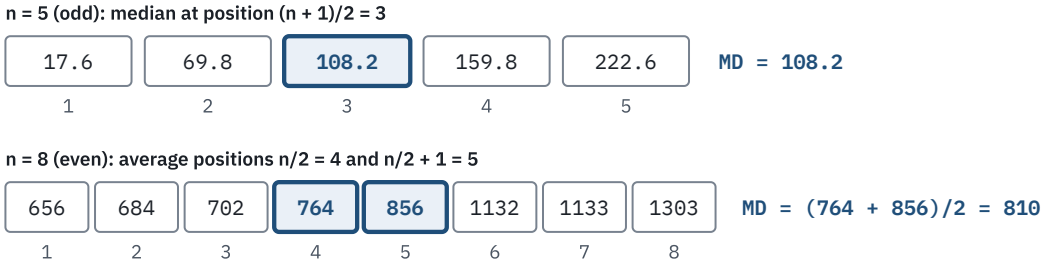
The mode and median for this table are worked out in parts 3 and 4 below; its variance is in 3-2.

الشرح بالعربي — متوسط البيانات المبوّبة

عندما تكون البيانات في جدول تكراري لا نعرف القيم الأصلية، نفترض أن كل قيم الفئة تساوي **مركزها** X_m (الحد الأدنى + الحد الأعلى ÷ 2). نضرب كل مركز في تكراره، نجمع، ثم نقسم على مجموع التكرارات: $\bar{x} = \Sigma f \cdot X_m / n$. انتبه: لو كانت هناك **فئة مفتوحة** (مثل «60 فأكثر») فلا يمكن حساب المتوسط لأن مركزها غير معروف، وعندها نستخدم الوسيط.

3 · The median (MD) BOTH

The **median** is the middle value of data that have been **arranged in order**. Steps: (1) order the data from smallest to largest; (2) count n; (3) if n is **odd**, the median is the middle value at position `(n + 1)/2`; if n is **even**, it is the mean of the values at positions `n/2` and `n/2 + 1`.



Order the data first. Odd n: the median is the single middle value (Example 3-4). Even n: it is the mean of the two middle values (Example 3-5). The number under each box is its position.

رتّب البيانات أولًا. إذا كان n فرديًا فالوسيط هو القيمة الوسطى (مثال 3-4، وإذا كان زوجيًا فهو متوسط القيمتين الوسطيين (مثال 3-5. الرقم تحت كل مربع هو ترتيبه.

INTERPRETING THE MEDIAN HADHRAMI

If the median exam score in a section is 76, then **50% of students scored 76 or less and 50% scored 76 or more.** The median is less sensitive to outliers and skewed data than the mean. BOTH

ACTIVITY · MEDIAN AMANI

Find the median of (a) 114, 77, 21, 101, 311, 77, 159, 382 and (b) 6180, 4894, 2821, 2559, 2483.

(a) Ordered: 21, 77, 77, 101, 114, 159, 311, 382. n = 8, so average the 4th and 5th values: `(101 + 114)/2 =` **107.5**

(b) Ordered: 2483, 2559, 2821, 4894, 6180. n = 5, so take the 3rd value: **2821**

Properties of the median: it finds the centre (middle value) of the data; it is used to tell whether a value falls in the upper or lower half; it **can** be used for an **open-ended** distribution; it is affected less than the mean by extremely high or low values. AMANI

MEDIAN FOR GROUPED DATA AMANI

(1) Find the total frequency N. (2) Calculate `N/2`. (3) Build the cumulative frequency (CF). The class whose CF **first exceeds** `N/2` is the **median class**.

Drivers: N = 71, so N/2 = 35.5. The CF column reads 19, **39**, ..., and 39 is the first CF above 35.5. The median class is therefore **29–36 (28.5–36.5)**. In words: the middle driver is aged between 29 and 36.

OPTIONAL: ONLY IF A QUESTION ASKS FOR ONE NUMBER

Interpolating inside the median class gives $L + ((N/2 - CF_{before})/f) \cdot w = 28.5 + ((35.5 - 19)/20) \cdot$ where L is the lower boundary, f the class frequency and w the class width. This formula is **not** in either deck. The course answer is the median class.

الشرح بالعربي — الوسيط

الوسيط هو **القيمة الوسطى بعد ترتيب البيانات**. إذا كان عدد القيم فرديًا نأخذ القيمة في الترتيب `(n+1)/2`، وإذا كان زوجيًا نأخذ متوسط القيمتين في الترتيبين `n/2` و `n/2 + 1`. أكثر خطأ شائع: **نسيان الترتيب قبل اختيار القيمة الوسطى**. ميزة الوسيط أنه لا يتأثر كثيرًا بالقيم المتطرفة ويمكن حسابه للفئات المفتوحة. وفي البيانات المبوّبة نحدد **فئة الوسيط**: أول فئة يتجاوز تكرارها المتجمع `N/2`.

4 · The mode BOTH

The **mode** is the value that occurs most often: the highest bar in a bar chart or histogram, sometimes described as the most typical case. It works for **qualitative and quantitative** data. Four cases:

- **Unimodal**: one value has the greatest frequency.
- **Bimodal**: two values tie for the greatest frequency, and both are modes.
- **Multimodal**: more than two values tie.
- **No mode**: no value occurs more than once. Never write "the mode is zero".

DATA	MODE
Ex 3-6 libraries: 21, 77, 77, 101, 114, 159, 311, 382	**77** · unimodal
Ex 3-7 nuclear reactors (15 years): 104 ×5, 107, 109 ×5, 110, 111 ×2, 112	**104 and 109** · bimodal
6180, 4894, 2821, 2559, 2483	**no mode**
Ex 3-9 CEO table (Ex 3-3)	modal class **15.5–20.5** (f = 13), midpoint 18
Drivers table	modal class **29–36** (f = 20), midpoint 32.5

For grouped data, the class with the highest frequency is the **modal class**, and its midpoint is sometimes reported as the mode. **Properties:** the mode is used when the most typical case is wanted; it is the easiest average to compute; it is the only average for nominal or categorical data (religious preference, gender, political affiliation); it is less sensitive to outliers. Textbook self-study: Ex 3-8 (U.S. patent leaders), Ex 3-10 (nonalcoholic beverages).

الشرح بالعربي — المنوال

المنوال هو **القيمة الأكثر تكرارًا**. قد يكون منوالًا واحدًا (unimodal)، أو اثنين (bimodal)، أو أكثر (multimodal)، أو **لا يوجد منوال** إذا لم تتكرر أي قيمة. احذر: لا تكتب «المنوال = صفر»، بل اكتب «لا يوجد منوال» (no mode). المنوال هو المقياس الوحيد الذي يصلح للبيانات **الوصفية الاسمية** مثل فصيلة الدم أو الجنس. في الجداول المبوّبة نحدد **الفئة المنوالية**، وهي الفئة ذات أعلى تكرار.

5 · The midrange (MR) HADHRAMI

MIDRANGE

$$MR = \frac{\text{lowest value} + \text{highest value}}{2}$$

A very rough estimate of the average. It uses only the two extremes, so a single outlier distorts it.

EX 3-12 · BANK FAILURES

3, 30, 148, 157, 71. Find the midrange, then compare it with the mean (in-class exercise).

$MR = (3 + 157)/2 =$ **80** $\quad \bar{x} = 409/5 =$ **81.8**

Here the two are close, but MR ignores the three middle values entirely.

6 · The weighted mean BOTH

The **weighted mean** accounts for the relative importance (weight) of each value. Use it when the values are **not equally represented**, that is, when some values contribute more to the average than others.

WEIGHTED MEAN

$$\bar{x} = \frac{w_1X_1 + w_2X_2 + \cdots + w_nX_n}{w_1 + w_2 + \cdots + w_n} = \frac{\Sigma wX}{\Sigma w}$$

EX 3-14 · GPA BOTH

A in English (3 credits), C in Psychology (3), B in Biology (4), D in PE (2). A = 4, B = 3, C = 2, D = 1, F = 0.

$$\bar{x} = \frac{3\cdot4 + 3\cdot2 + 4\cdot3 + 2\cdot1}{3 + 3 + 4 + 2} = \frac{32}{12} = 2.667 \approx \mathbf{2.7}$$

ACTIVITY · COURSE GRADE AMANI

Homework 10%, quizzes 15%, midterm 25%, final 50%. Homework 40, 50; quizzes 50, 60, 20, 50, 40; midterm 65; final 95. Estimate the weighted mean.

HW 27 · FINAL GRADE HADHRAMI

Exams 20%, term paper 30%, final 50%. Grades 83, 72, 90.

Average each component first: homework `45`, quizzes `220/5 = 44`.

$$0.10(45) + 0.15(44) + 0.25(65) + 0.50(95)$$

$$= 4.5 + 6.6 + 16.25 + 47.5 = \boxed{74.85}$$

The weights already sum to 1 (100%), so no further division is needed.

$$0.2(83) + 0.3(72) + 0.5(90) = \boxed{83.2}$$

HW 24 · FAT GRAMS PER OUNCE HADHRAMI

3 oz shrimp 3.33 g/oz · 3 oz veal 3.00 · 2 oz beef 2.50 · 2.5 oz chicken 4.40 · 4 oz tuna 1.75.

The weights are the ounces eaten.

$$\Sigma wX = 9.99 + 9 + 5 + 11 + 7 = 41.99;\ \Sigma w = 14.5$$

$$\bar{x} = 41.99/14.5 = \boxed{2.90\ \text{g/oz}}$$

الشرح بالعربي — المتوسط المرجّح

نستخدمه عندما **لا تكون كل القيم بنفس الأهمية**، مثل حساب المعدل التراكمي (كل مادة لها عدد ساعات مختلف)، أو الدرجة النهائية (لكل تقييم نسبة مختلفة). نضرب كل قيمة في وزنها، نجمع، ثم نقسم على **مجموع الأوزان**. إذا كانت الأوزان نسبًا مجموعها 100% فلا نقسم على شيء. خطأ شائع: نسيان أخذ متوسط مكوّنات البند الواحد أولًا (مثل متوسط الكويزات الخمسة) قبل ضربها في الوزن.

7 · Which measure should you report? BOTH · CLO3-C

LEVEL OF MEASUREMENT	POSSIBLE	BEST
Nominal	mode	**mode**
Ordinal	median, mode	**median**
Interval / ratio, not skewed	mean, median, mode	**mean**
Interval / ratio, skewed or with outliers	mean, median, mode	**median**

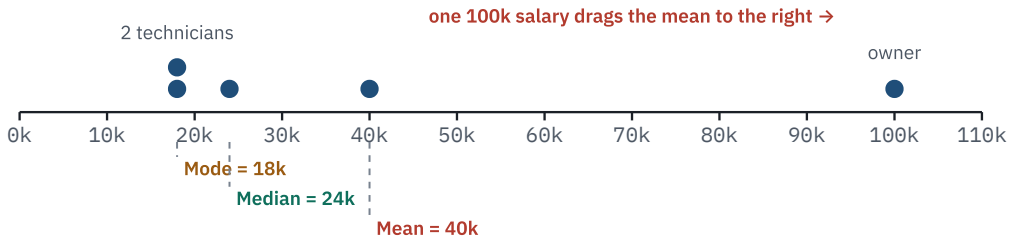
TABLE 3-1 · SUMMARY HADHRAMI

MEASURE	DEFINITION	SYMBOL
Mean المتوسط الحسابي	Sum of values ÷ number of values	μ, $\bar{x}$
Median الوسيط	Middle point of ordered data	MD
Mode المنوال	Most frequent data value	none
Midrange منتصف المدى	(lowest + highest)/2	MR

EX 3-11 · SALARIES OF PERSONNEL (THE ENTIRE POPULATION) HADHRAMI

Owner $100,000 · manager 40,000 · salesperson 24,000 · two technicians 18,000 each. Find the mean, median and mode. Which best represents the company?

μ = 200,000/5 = **$40,000** MD = **$24,000** mode = **$18,000**

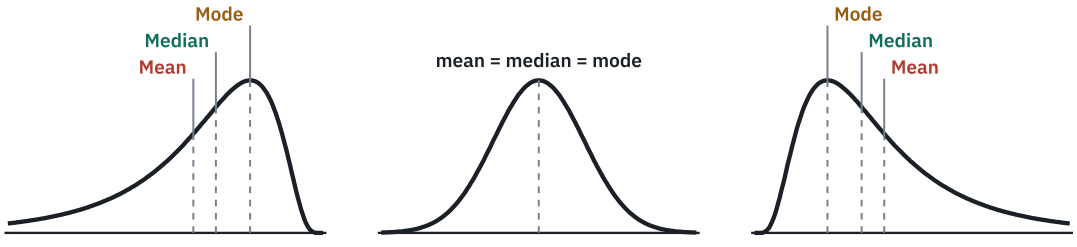


Example 3-11. One extreme salary pulls the mean up to 40k while the median (24k) and mode (18k) stay with the typical workers. With outliers or skew, report the median.

مثال 3-11: راتب واحد كبير (100 ألف) يسحب المتوسط إلى 40 ألفًا، بينما يبقى الوسيط (24 ألفًا) والمنوال (18 ألفًا) قريبين من رواتب أغلب العاملين. مع القيم المتطرفة أو الالتواء نستخدم الوسيط.

The owner's extreme salary pulls the mean far above the median and mode. The **median** is the more representative measure here.

8 · Distribution shape BOTH · CLO3-B



Negatively (left) skewed
`mean < median < mode`

Symmetric
`mean = median = mode`

Positively (right) skewed
`mode < median < mean`

The mean is pulled toward the long tail; the mode stays at the peak; the median sits between them. Read the shape from the order of the three measures.

المتوسط ينسحب نحو الذيل الطويل، والمنوال يبقى عند القمة، والوسيط بينهما. من ترتيب المقاييس الثلاثة نعرف شكل التوزيع.

HOMEWORK · STEM-AND-LEAF (KEY 1 | 0 = 10) AMANI

1 | 0 2 3 5 5 5 6 7 7 9 · 2 | 0 2 3 · 3 | · 4 | 5. Find the mean, median and mode, and describe the shape.

As plotted (n = 14, ΣX = 259): `x̄ = 18.5`; median = mean of the 7th and 8th values = `(16 + 17)/2 = 16.5`; mode = `15`.

Printed answers (19.6, 17, 15) correspond to one extra value, 35 (n = 15, ΣX = 294).

In both versions mode < median < mean: **`positively (right) skewed`**. The value 45 forms the long right tail.

ACTIVE LEARNING · STEM-AND-LEAF AMANI

1 | 2 3 4 · 2 | 3 4 5 6 7 · 3 | 0 1 2 · 4 | 2 4 5 6 7. (1) Minimum and maximum. (2) Mean, median, mode. (3) Shape.

(1) min = **12**, max = **47**. (2) n = 16, ΣX = 481, so `x̄ = 30.06`; median = `(27 + 30)/2 = 28.5`; **`no mode`**, since every value appears once.

(3) The mean is only slightly above the median: the distribution is roughly symmetric with a slight right skew. With no mode, compare the mean and median.

الشرح بالعربي — شكل التوزيع

نعرف الشكل من **ترتيب المقاييس الثلاثة**. إذا كان المنوال < الوسيط < المتوسط فالتوزيع **ملتوٍ لليمين** (موجب)، لأن الذيل الطويل يمين والمتوسط ينسحب نحوه. إذا كان المتوسط < الوسيط < المنوال فهو **ملتوٍ لليسار** (سالب). وإذا تساوت الثلاثة فهو **متماثل**. قاعدة سريعة: **المتوسط يتبع الذيل**. وإذا لم يوجد منوال فقارن المتوسط بالوسيط فقط.

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · **01 · Lesson 3-1: Measures of Central Tendency** · 02 · Lesson 3-2: Measures of Variation · 03 · Lesson 3-3: Measures of Position · 04 · Lesson 3-4: Exploratory Data Analysis · 05 · Revision pack: formulas, decisions, traps, R · 06 · Exam-style questions (no answers) · 07 · Full worked answers